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STAFF REPORT FOR THE 2022 ARTICLE IV CONSULTATION AND REQUESTS FOR AN EXTENDED ARRANGEMENT UNDER THE EXTENDED FUND FACILITY AND AN ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

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Benin Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Moderate*[1]
Overall risk of debt distress	*Moderate*
Granularity in the risk rating	*Limited space to absorb shocks*
Application of judgment	*No*

Benin remains at moderate risk of external and overall debt distress, unchanged from the previous DSA (December 2020). All the projected external debt burden indicators are below their thresholds under the baseline scenario. Nevertheless, the space to absorb shocks remains limited, with external debt burden indicators breaching high-risk thresholds in selected stress scenarios, in particular related to exports, which illustrates vulnerabilities related to Benin's transit-dependent economy. The high debt service to revenue ratio leaves debt vulnerable to revenue underperformance or shifts in market sentiment that could increase rollover costs. Moreover, although the baseline present value (PV) of public debt-to-GDP ratio remains below its prudent benchmark, it is vulnerable to natural disasters as illustrated by the shock scenario, which was applied based on Benin's historic and potential climate-related exposure to natural disaster shocks. Sustained revenue mobilization efforts, including in the context of the IMF-supported program, along with continued prudent borrowing and active debt management strategy, will be important for mitigating the risk of debt distress.

[1] Prepared by the IMF and the World Bank. This DSA follows the Guidance Note of the Join Bank-Fund Debt Sustainability Framework for Low Income Countries, February 2018.

PUBLIC DEBT COVERAGE

1. This Debt Sustainability Analysis (DSA) covers central government and central bank debt, as well as guarantees provided by the central government (Text Table 1).[2] Central bank debt borrowed on behalf of the government (i.e., debt to the IMF) is included as external debt. External debt is defined on a currency basis owing to data limitations, except for debt from the regional development bank (BOAD), which is included as external debt for the purpose of the DSA. The baseline also includes on-lending from the central bank related to the 2021 IMF general SDR allocation as part of the domestic debt stock: in line with the WAEMU-wide agreement, the BCEAO on-lent Benin its portion of the SDR allocation (1.0 percent of GDP).

Text Table 1. Benin: Public Debt Coverage Under the Baseline Scenario

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

Text Table 2. Benin: Contingent Liability Stress Test

1	The country's coverage of public debt	The central government, central bank, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	1.4	Stock of SOE debt not captured in the baslined (based on end-2021 estimate).
4	PPP	35 percent of PPP stock	2.4	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		8.8	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND

A. Recent Debt Developments

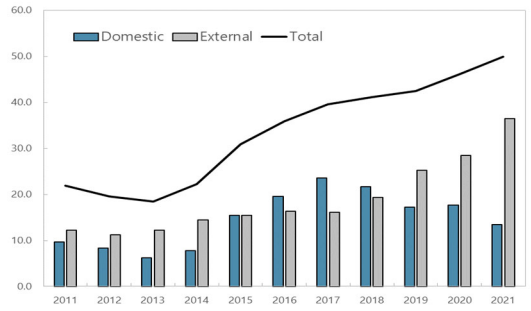
2. Benin's public debt continued to increase in 2021. Debt has been on an upward trend since 2014, driven by the scale-up in public investment, the large COVID-19-driven deficits in 2020-

[2] Debt on-lent to SOEs is also included as part of central government borrowing.

2021. Debt rose from 22 percent of GDP in 2014 to 49.9 percent of GDP in 2021.[3] In 2021, Benin also took advantage of favorable market conditions to issue a €1 billion Eurobond (6.8 percent of GDP), which it used for investment and to finance nearly two-thirds of the Eurobond coming due in 2026 to mitigate rollover risk.

3. Proactive liability management has helped smooth out the public debt service profile and reduce costs. The overall public debt service to revenue ratio is expected to average around 34 percent in 2022-26, compared with 64 percent projected in the previous DSA (December 2020). The reimbursement of around 65 percent of the 2019 Eurobond falling due in 2024–26 helped smooth the amortization profile by a cumulative 2 percent of GDP and reduced 2022-26 interest costs by 0.4 percent of GDP. This follows earlier operations such as the issuance of a US$300 million external commercial loan in 2018 backed by a World Bank partial guarantee to reprofile costly short term-domestic debt.

Text Figure 1. Benin: Total Public Debt
2011-21
(Percent of GDP)



Sources: Country authorities; and staff estimates and projections

4. The composition of debt has shifted gradually towards external debt in recent years, where borrowing costs were comparatively lower (Text Table 3). The largest shares of debt at end-2021 were held by multilateral creditors and international bond holders at 37 percent and 22 percent, respectively. Domestic debt made up around 27 percent of the debt stock, mostly government securities on the regional financial market. While maturities have typically been subject to shorter-term rollover periods compared with external debt, the authorities have started to issue domestic debt with longer maturities, and rollover risks are further mitigated by the liquidity and relative stability of the regional market.

5. Data limitations prevent the inclusion of non-guaranteed SOE debt in the baseline, which is captured in the contingent liability stress test. The authorities have made progress on this front in recent years, collecting information to produce an estimate of commercial debt for all 13 non-guaranteed SOEs, which stood at 1.4 at end-2021. The authorities have also agreed to publish annually SOE debt outstanding, including commercial debt, and include details on on-lending to SOEs in quarterly debt bulletins, as part of IDA's Sustainable Development Finance Policy (SDFP) and most recent Development Policy Operation.[4] In 2021, they started including details on on-lending to SOEs in quarterly debt bulletins. Also, under the SDFP, the Debt Management Office and the Directorate in charge of SOEs (DGPED) established a monitoring system following the adoption by ministerial order of a risk-based framework for granting SOE guarantees.

[3] Benin does not currently have any arrears vis-à-vis external creditors. Public domestic debt arrears are commitments to domestic suppliers, hence would not trigger a debt distress event.

[4] Since Q3 2021, the authorities have started to publish the updated debt bulletins on the dedicated website.

6. Nevertheless, more information on the finances of the SOEs (e.g., revenues) and their debt is needed to fully incorporate them into the baseline. To reflect risks associated with this debt not captured in the baseline, the contingent liabilities shock is calibrated to 8.8 percent of GDP, which includes 1.4 GDP percent for SOE debt, in addition to default settings for PPPs (2.4 percent of GDP) and financial market risk (5.0 percent of GDP) (Text Table 2). Benin's debt management capacity is high as reflected in the 2020 and 2021 CPIA evaluations.

Table 1. Benin: Decomposition of Public Debt and Debt Service by Creditor, 2021-23[1]

	Debt Stock (end of period)			Debt Service					
	2021			2021	2022	2023	2021	2022	2023
	(In US$)	*(Percent total debt)*	*(Percent GDP)*	*(In US$)*			*(Percent GDP)*		
Total	8441.7	100	49.9	2248	855	970	13.3	4.6	4.8
External	6162.3	73.0	36.5	690	325	363	4.084	1.75	1.79
Multilateral creditors[2]	3128	37.1	18.5	124	120	136	0.733	0.65	0.67
IMF	450	5.3	2.7						
World Bank	1318	15.6	7.8						
ADB/AfDB/IADB	472	5.6	2.8						
Other Multilaterals	888	10.5	5.3						
Arab Bank for Economic Development	54	0.6	0.3						
BOAD	376	4.4	2.2						
Nordic Development Fund	15	0.2	0.1						
ECOWAS Bank for Investment and Development	68	0.8	0.4						
European Investment Bank	64	0.8	0.4						
IFAD	63	0.7	0.4						
OPEC	29	0.3	0.2						
Islamic Development Bank	212	2.5	1.3						
Bilateral Creditors	478	5.7	2.8	81	33	36	0.5	0.2	0.2
Paris Club	57	0.7	0.3	0	2	3	0.0	0.0	0.0
France	57	0.7	0.3						
Non-Paris Club	421	5.0	2.5	81	31	34	0.5	0.2	0.2
China	323	3.8	1.9						
India	19	0.2	0.1						
Kuwait	40	0.5	0.2						
Saudi Arabia	37	0.4	0.2						
Bonds	1895	22.4	11.2	485	95	102	2.9	0.5	0.5
Commercial creditors	662	7.8	3.9	0	77	89	0.0	0.4	0.4
MUFG Bank	254	3.0	1.5						
RABOBANK	164	1.9	1.0						
Bank of China	12	0.1	0.1						
Societe General	59	0.7	0.3						
UKEF	46	0.5	0.3						
Banco de Brazil	55	0.6	0.3						
Deutche Bank	36	0.4	0.2						
NTXS	28	0.3	0.2						
BPI France	7	0.1	0.0						
Credit Suisse	3	0.0	0.0						
Domestic	2279	27.0	13.5	1557	529	608	9.2	2.9	3.0
Held by residents, total	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Held by non-residents, total	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
T-Bills	0	0.0	0.0						
Bonds	2001	23.7	11.8						
Loans[3]	278	3.3	1.6						
Memo items:									
Collateralized debt	0		0.0						
Contingent liabilities	233		1.4						
o/w: Public guarantees	0		0.0						
o/w: Other explicit contingent liabilities[4]	233		1.4						
Nominal GDP			16905	16904.6	18556	20247			

1/As reported by Country authorities according to their classification of creditors, including by official and commercial.

2/Multilateral creditors" are simply institutions with more than one official shareholder and may not necessarily align with creditor classification under other IMF policies (e.g. Lending Into Arrears).

3/Includes central bank on lending related to the SDR allocation.

4/Estimation of commerical non-guaranteed SOE debt that is not included in the debt stock based on end-2021 estimation.

Sources: Country authorities; and staff estimates and projections

B. Macroeconomic Assumptions

7. Macroeconomic assumptions underlying the DSA projections are consistent with the program baseline (Text Table 4). Compared with the previous DSA (at the time of Fund emergency financing under the RCF/RFI in December 2020), growth in 2021 was more robust given the economy's higher-than-anticipated resilience to the dual COVID-19 and Nigeria border closure shocks.[5] Meanwhile in projection years there is a slight mark down in 2022 growth due to the Russia-Ukraine shock and somewhat lower growth rates in the medium to long-term based on updated potential growth estimates. Primary deficits are higher, reflecting fiscal accommodation for priority spending needs, which are partly offset in part by revenue mobilization measures implemented in the context of the IMF-supported program. Current account deficits are also wider reflecting the impact of the war in Ukraine (terms of trade shock) and higher government expenditure, particularly security-related. The main macroeconomic assumptions are as follows:

- **Real GDP Growth.** Although the negative terms of trade shock to the war in Ukraine will take a toll on growth, economic activity will be supported by public investment and robust port activity in 2022, with growth forecasted at 5.7 percent. This is only slightly below the previous DSA projection (6.0 percent), reflecting the economy's higher-than-anticipated resilience to the to the COVID-19 and Nigeria border closure shocks. Over the medium-term, large-infrastructure investment consistent with the authorities' development objectives[6] and some recovery in private investment bolstered by the authorities' efforts (e.g., acceleration of the Special Economic Zone, promotion of SMEs) will continue to support growth, which is expected to remain around potential at 6 percent.[7] Similarly, longer-term projections remain conservative at 5.7 percent converging to the steady-state.

- **Inflation.** While inflation is projected to reach 5 percent in 2022, driven by higher food and oil prices; it is expected to converge to around 2 percent over the medium-term, consistent with the peg to the euro.

- **Primary fiscal balance.** The primary deficit (including grants) is expected to widen to -3.7 percent of GDP in 2022, compared with -1.0 percent during the previous DSA, reflecting appropriate fiscal loosening in 2020 and 2021 to accommodate priority spending needs, unanticipated spending to mitigate security-related risks, hiring in the health sector, and wage increases to help offset some cumulative purchasing power erosion. Sustained domestic revenue

[5] Significant strides in macroeconomic management, supported by the 2017-20 ECF put the economy on solid footing to sustain the dual shocks (See *Benin : Sixth Review under the Extended Credit Facility Arrangement, and Request for Augmentation of Access* (IMF, 2020)).

[6] Benin's National Development Plan (PND; 2018-25) emphasizes Sustainable Development Goals (SDG), particularly closing infrastructure and human capital gaps by scaling up spending on education, health, access to water, and electricity. The IMF 2022-25 ECF/EFF will help anchor this development plan by focusing on creating fiscal space to support significant development needs while preserving debt sustainability.

[7] Estimate is based on a growth accounting exercise, using envisaged public and private investment dynamics for 2023-26, average historical contributions to growth of human capital accumulation for 2015-18 and estimated total factor productivity during the previous IMF-supported ECF (2017-20).

mobilization efforts in the context of the Fund-supported program[8], starting with a package of tax policy and administrative measures in 2022 totaling 0.5 ppt of GDP and further supported by the authorities' planned Medium-Term Revenue Strategy, will help raise Benin's relatively low tax-to-GDP ratio. Higher revenues together with spending prioritization, efficiency gains in public investment, unwinding of temporary COVID-19-related spending (as the pandemic wanes), and interest bill savings thanks to active debt management, will support convergence to WAEMU current fiscal norms (overall balance of 3 percent of GDP by 2024 and a wage-to-tax revenue ratio of 35 percent).

- **Current account deficit.** The current account deficit is expected to widen to about 6 percent of GDP in 2022 driven by higher food and oil prices and increased government spending including security-related outlays (with high import content) and the persistent terms of trade shock amid the war in Ukraine. It is expected to improve over the medium and long term, supported by fiscal consolidation and ongoing reforms to boost competitiveness offsetting projected declines in cotton export receipts as international prices moderate.

8. Consistent with the authorities' borrowing plan, this DSA assumes that the authorities will continue to maintain a prudent borrowing strategy, maximizing concessional resources to the extent possible. The baseline includes financing from a potential IMF-supported program as well as financing from the World Bank in line with IDA allocations. Remaining external needs in the medium-term would be filled on largely semi-concessional terms by other multilateral and bilateral creditors, except for a projected US$500 million Eurobond issuance consistent with the authorities' desire to maintain market access.[9] Grant-equivalent financing is expected to steadily decline as a share of GDP as Benin's relative income increases. Net external official financing flows would average 1.6 percent of GDP over the projection period, following a declining trend. Over time, Benin is also expected to rely more on domestic sources of financing as the domestic debt market continues to develop.[10]

9. Public debt is projected to decline over the medium to long term, as a result of prudent fiscal policy and steady growth. After peaking at 52 percent of GDP at end-2023, reflecting fiscal policy accommodation to contain the socio-economic fallout from the protracted COVID-19 pandemic, the war in Ukraine and mitigation of security risks. Public debt is projected to decline to 44 percent by 2032 as growth converges to its potential. The debt trajectory is predicated on the authorities implementing a revenue-based fiscal consolidation to ensure convergence to WAEMU regional fiscal norms.

[8] In addition to an indicative target on tax revenue, the Fund-supported program will have structural benchmarks related to the Medium-Term Revenue Strategy and tax expenditures to support the authorities' revenue mobilization efforts.

[9] The grant element of new borrowing over the projection period would average 26 percent.

[10] This DSA assumes domestic financing will come mostly in the form of one-to-seven-year bonds with a small portion of longer maturities at rates from 6 to 7 percent, with some scope for upside risks given recent issuance experience.

10. Macro-fiscal projections are realistic (Figures 3-4). The three-year primary adjustment is consistent with historical cross-country data, falling just below the top quartile for historic adjustments in LICs with IMF-supported programs. The growth path is does not exceed those derived from typical fiscal multipliers for LICs. Public and private investment rates are similar to the previous DSA as is the contribution of public investment to growth, which is modest. While drivers of debt indicate large residual components contributing to unanticipated changes in debt over the past five years, this is consistent with the unanticipated nature of the COVID-19 shock.[11]

11. The macroeconomic outlook is subject to a number of risks, which remain tilted to the downside. Like other countries in the region, Benin remains susceptible to further COVID outbreaks owing to a still low vaccination rate. A contagion of security risks could weigh further on the budget. Further increases in food and oil prices amid ongoing geopolitical tensions related to the war in Ukraine would weigh on the current account and could fuel social discontent. A shift in global risk appetite could complicate medium-term rollover risks. A further increase in international food and oil prices—amidst ongoing war in Ukraine—would exert pressure on external accounts while further eroding the purchasing power of the vulnerable and fueling social tensions. Over the long-term, the relatively small size of the economy could leave the economy more vulnerable to shocks, posing risks to growth.

C. Country Classification and Determination of Stress Test Scenarios

12. Benin's debt carrying capacity continues to be classified as medium. Based on a calculation of a composite indicator reflecting factors such as the 2020 WB CPIA index, and October 2021 WEO (real growth rates, reserve coverage, remittances, and world growth), Benin has a CI score of 3.00. (Text Table 5). As a result, this DSA continues to use the same external debt burden thresholds and total public benchmarks for countries with medium debt carrying capacity (Text Table 6).

13. Stress tests generally follow standardized settings, with the addition of tailored stress tests to capture risks related to contingent liabilities, natural disasters, commodity prices, and market financing. Given Benin's high vulnerability to natural disasters and vulnerability to climate change (particularly from flooding and coastal erosion), the standard natural disaster shock has been applied. Commodity exports (cotton) make up a significant part of the export base leaving it open to potential price shocks. Finally, outstanding Eurobonds may leave Benin exposed to rollover risk in the event of a change in global risk sentiment—though the current maturity profile mitigates this risk over the next two years.

[11] The domestic debt stock includes debt flows in the form of COVID-19 economic support measures (e.g., credit lines and guarantees) that were not part of the fiscal deficit.

Text Table 4. Benin: Baseline Macroeconomic Assumptions for Debt Sustainability Analysis

	2021	2022	2023	2024	2025	2026	Medium-term 2022-26	Long-term 2027-42
GDP Growth (percent)								
Current DSA	7.2	5.7	6.2	6.0	6.0	6.0	6.0	5.7
Previous DSA[1]	5.0	6.0	6.5	6.5	6.5	6.4	6.4	6.5
GDP Deflator (percent)								
Current DSA	1.6	4.9	1.0	1.5	2.0	2.0	2.3	2.0
Previous DSA	3.7	7.7	1.4	-0.2	0.9	0.7	2.1	1.5
Current account deficit (percent GDP)								
Current DSA	-4.4	-6.2	-5.7	-4.6	-5.0	-4.5	-5.2	-4.1
Previous DSA	-3.9	-4.1	-3.8	-3.9	-3.8	-3.8	-3.9	-3.8
Exports								
Current DSA	24.0	25.3	23.9	23.0	22.8	22.6	23.5	22.4
Previous DSA	21.7	23.3	23.0	22.9	22.8	22.8	22.9	22.8
Primary Deficit (percent GDP)								
Current DSA	-3.5	-3.7	-2.8	-1.3	-1.3	-1.3	-2.1	-1.4
Previous DSA	-2.1	-1.0	-0.5	-0.1	-0.2	0.4	-0.3	0.1
Revenue and grants (percent GDP)								
Current DSA	14.1	14.3	15.0	15.5	15.9	16.2	15.4	19.1
Previous DSA	12.7	13.1	13.7	13.9	13.9	14.1	13.7	15.3

Sources: IMF Staff estimates and projections.

1/ December 2020 RCF/RFI Request

Text Table 5. Benin: Calculation of Debt-Carrying Capacity

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.579	1.38	46%
Real growth rate (in percent)	2.719	5.791	0.16	5%
Import coverage of reserves (in percent)	4.052	45.974	1.86	62%
Import coverage of reserves^2 (in percent)	-3.990	21.136	-0.84	-28%
Remittances (in percent)	2.022	0.808	0.02	1%
World economic growth (in percent)	13.520	3.137	0.42	14%
CI Score			3.00	100%
CI rating			Medium	

DEBT SUSTAINABILITY RESULTS

A. External Debt Sustainability

14. All the external debt burden indicators remain below their policy-dependent thresholds (Table 1, Figure 1). This represents an improvement from the previous DSA where debt-service to revenue ratio exceeded its threshold in 2024-25 (19.1-19.9 percent) owing to the bullet Eurobond repayment. The authorities have since smoothed out this repayment thanks to the 2021 liability management operations. The PV of total PPG external debt to GDP is expected to remain well below the threshold throughout the projection period and follow a steadily declining path, averaging 30 percent of GDP in 2022-26 and 25 percent of GDP in the long-term.

Text Table 6. Benin: Debt Burden Thresholds & Benchmarks

EXTERNAL debt burden thresholds	Weak	Medium	Strong
PV of debt in % of			
Exports	140	180	240
GDP	30	40	55
Debt service in % of			
Exports	10	15	21
Revenue	14	18	23

TOTAL public debt benchmark	Weak	Medium	Strong
PV of total public debt in percent of GDP	35	55	70

15. Stress tests highlight Benin's vulnerability to shocks, particularly those related to commodity prices and exports. Three out of the four debt-burden indicators breach their thresholds under certain stress tests: the PV of debt-to-exports and debt service-to-exports ratio where the export related shock[12] is the most extreme shock; and for the debt service-to-revenue ratio, where a one-time nominal currency depreciation against the USD is the most extreme shock[13], and a shock to exports would also cause several breaches. These three debt burden indicators would also be breached under a commodity price shock scenario.[14] While the PV of debt-to-GDP ratio does not breach thresholds under any stress tests, a shock to exports would be the most extreme shock. These shocks illustrate the risks posed by limited economic diversification, with the Beninese economy centered on transit trade and highly dependent on economic developments in Nigeria (See Staff Report Annex VI). Although the historical scenario is relatively more severe than the baseline, the calibration period of 2012–2021 captures particularly severe periods in Benin's economy including the impact of COVID-19, the Nigeria border-closure, and the 2015 downturn. Moreover, compared with the historical period continued efforts to mobilize revenues and active debt management are expected to help stabilize debt levels.

16. The granularity assessment indicates that Benin has limited space to absorb shocks (Figure 5). Under the module, which allows qualifying the moderate risk of debt-distress, Benin is assessed as having limited space because the baseline debt-service-to-revenue indicator is close enough to its respective threshold that a median observed shock would result in a downgrade to high-risk.

[12]Based on the standard stress test settings, this export stress test applies a nominal export shock applies a 14 percent reduction in nominal export growth of 14 percent in 2023–24. Interactions with real GDP growth are also included.

[13] Based on standard stress test settings, this test applies a 30 percent nominal depreciation of the FCFA (which is pegged to the euro) against the USD against the USD in 2023. Interactions with inflation are also included.

[14] Based on the standard stress settings, this commodity price stress test applies a commodity price gap of 21 percent price shock in 2023, which closes over 6 years. Interactions for real GDP, revenue-to-GDP, and the GDP deflator are also included.

17. The market-financing module suggests that market risks are moderate (Figure 6). Although EMBI spreads are above the benchmark and increasing reflecting the recent volatility bout amid the war in Ukraine, gross financing needs remain well below the respective benchmark and potential for heightened liquidity needs is moderate. The debt-service to revenue ratios would exceed their thresholds in a market financing shock scenario in 2026 and 2030, given the repayment profile discussed above (which by design excludes possible liability management operations that would help manage these risks).

B. Total Public Debt Sustainability

18. Total public PPG remains below its respective benchmark in the baseline (Figure 2 and Table 2). The present value of public debt averages 40 percent of GDP over the projection period, well below the 55 percent benchmark.

19. Stress tests indicate that public debt is most vulnerable to commodity price shocks and natural disasters. For the PV of debt-to-GDP ratio, a commodity price shock would be the most extreme shock, resulting in a breach of the benchmark in 2029–2032. Under the natural disaster scenario (which simulates a one-off real GDP shock of around 2 percent in 2023), the PV of debt-to-GDP would breach the benchmark in 2023, with large increases in debt-to-revenue and debt service-to-revenue ratios (the latter exceeding 60 percent in 2027).

RISK RATING AND VULNERABILITIES

20. This DSA finds that Benin continues to remain at moderate risk of external and overall debt distress, unchanged from the previous DSA. All external debt indicators remain below their high-risk thresholds and benchmarks. Although Benin's debt levels appear relatively manageable, these levels remain vulnerable to shocks, particularly those related to exports, currency depreciation, and natural disasters. As evidenced by the high debt service to revenues ratio, large amortization payments and the low revenue base may pose liquidity risks. Lower than expected revenues, including due to policy implementation lags, or a shift in market sentiment leading to higher borrowing and rollover costs could heighten debt risks.

21. Sustained revenue mobilization efforts along with a prudent borrowing and debt management strategy will be important for mitigating the risk of debt distress. As highlighted by the granularity assessment, Benin has limited space to absorb shocks owing to the narrow distance between the debt service to revenue ratio and high-risk threshold in years where large Eurobond bullet repayments are due. Continued proactive liability management to facilitate the rollover of these payments, maximizing concessional borrowing, and measures to bring tax revenues closer to Benin's potential (**SIP–II on domestic revenue mobilization**) will help mitigate risks to debt distress. Fiscal policy guided by an interest to revenue ratio of 19 percent as envisaged under the IMF-supported program, and convergence to the WAEMU fiscal deficit norm of 3 percent of GDP by 2024, can help guide these policies.

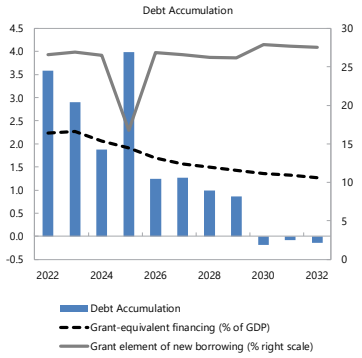
22. The authorities broadly concurred with the moderate risk of external debt distress rating. They are of the view that debt from BOAD should be classified as domestic debt, citing the institution's governance structure and the regional arrangement that makes the CFA currency union a consolidated unit. The authorities also noted that classifying such debt as domestic would further improve the external debt profile. They believe that some of the financing assumptions, particularly the terms of market access, are conservative. The authorities remain committed to preserving debt sustainability, mitigating refinancing risks, and containing borrowing costs through a combination of revenue mobilization, active debt management, and prudent borrowing.

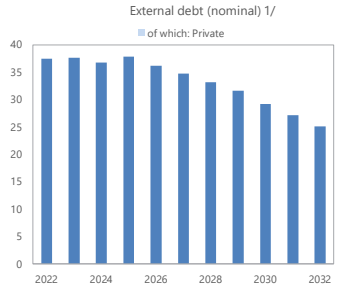
Table 1. Benin: External Debt Sustainability Framework, Baseline Scenario, 2019-2042

(in percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
External debt (nominal) 1/	**25.3**	**28.4**	**36.5**	**37.5**	**37.6**	**36.8**	**37.8**	**36.2**	**34.7**	**25.1**	**13.6**	**19.5**	**33.3**
of which: public and publicly guaranteed (PPG)	**25.3**	**28.4**	**36.5**	**37.5**	**37.6**	**36.8**	**37.8**	**36.2**	**34.7**	**25.1**	**13.6**	**19.5**	**33.3**
Change in external debt	5.9	3.2	8.0	1.0	0.1	-0.8	1.0	-1.6	-1.5	-2.0	-0.8		
Identified net debt-creating flows	**5.1**	**0.7**	**2.4**	**5.4**	**4.7**	**3.7**	**4.0**	**3.6**	**3.2**	**3.8**	**4.5**	**5.2**	**3.8**
Non-interest current account deficit	**3.6**	**1.3**	**3.7**	**5.4**	**4.8**	**3.8**	**4.1**	**3.6**	**3.2**	**3.4**	**3.8**	**4.2**	**3.8**
Deficit in balance of goods and services	5.0	2.8	4.7	6.9	6.4	5.4	5.7	5.3	5.0	5.0	5.0	**5.7**	**5.4**
Exports	24.9	22.4	24.0	25.3	23.9	23.0	22.8	22.6	22.4	22.4	22.4		
Imports	29.9	25.1	28.7	32.2	30.3	28.5	28.5	27.9	27.5	27.5	27.5		
Net current transfers (negative = inflow)	-1.3	-1.8	-1.2	-1.3	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4	-1.4	**-2.1**	**-1.4**
of which: official	-0.4	-0.7	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1		
Other current account flows (negative = net inflow)	-0.1	0.2	0.1	-0.2	-0.3	-0.3	-0.2	-0.3	-0.4	-0.1	0.2	**0.6**	**-0.3**
Net FDI (negative = inflow)	**1.3**	**1.0**	**1.2**	**1.2**	**1.1**	**1.1**	**1.1**	**1.1**	**1.1**	**1.1**	**1.1**	**1.6**	**1.1**
Endogenous debt dynamics 2/	**0.3**	**-1.6**	**-2.5**	**-1.2**	**-1.3**	**-1.2**	**-1.2**	**-1.2**	**-1.1**	**-0.8**	**-0.4**	**-0.6**	**-1.1**
Contribution from nominal interest rate	0.4	0.5	0.7	0.8	0.9	0.9	0.8	0.9	0.9	0.7	0.3	**0.3**	**0.8**
Contribution from real GDP growth	-1.3	-0.9	-1.8	-2.0	-2.1	-2.1	-2.0	-2.1	-2.0	-1.5	-0.7	**-0.9**	**-1.9**
Contribution from price and exchange rate changes	1.1	-1.2	-1.5	...	...	...	...	...	...	...	...	**-0.03**	**...**
Residual 3/	**0.8**	**2.5**	**5.6**	**-4.4**	**-4.5**	**-4.5**	**-3.0**	**-5.2**	**-4.7**	**-5.8**	**-5.3**	**-2.7**	**-4.9**
of which: exceptional financing	0.0	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**28.4**	**30.7**	**30.8**	**30.0**	**31.2**	**29.8**	**28.6**	**20.4**	**10.9**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**118.6**	**121.3**	**129.2**	**130.4**	**136.9**	**131.9**	**127.4**	**90.8**	**48.6**		
PPG debt service-to-exports ratio	**4.9**	**5.5**	**8.5**	**7.0**	**8.0**	**10.8**	**10.9**	**10.4**	**9.2**	**10.9**	**4.8**		
PPG debt service-to-revenue ratio	**9.4**	**9.7**	**15.5**	**13.2**	**13.7**	**17.2**	**16.7**	**15.4**	**13.2**	**13.9**	**6.1**		
Gross external financing need (Billion of U.S. dollars)	0.9	0.5	1.2	1.5	1.6	1.6	1.8	1.8	1.8	2.9	5.2		
Key macroeconomic assumptions													
Real GDP growth (in percent)	6.9	3.8	7.2	5.7	6.2	6.0	6.0	6.0	6.0	5.9	5.4	**5.4**	**6.0**
GDP deflator in US dollar terms (change in percent)	-5.5	4.9	5.4	-1.3	2.5	2.8	2.9	2.7	2.4	2.0	2.0	**0.0**	**2.0**
Effective interest rate (percent) 4/	2.3	2.1	3.0	2.3	2.5	2.5	2.5	2.7	2.7	2.7	2.4	**1.8**	**2.6**
Growth of exports of G&S (US dollar terms, in percent)	-6.9	-2.3	21.1	10.1	2.7	5.2	7.9	8.0	7.9	8.0	7.5	**10.6**	**7.4**
Growth of imports of G&S (US dollar terms, in percent)	-7.8	-8.6	29.1	17.1	2.4	2.3	9.1	6.7	6.8	8.0	7.5	**9.3**	**7.7**
Grant element of new public sector borrowing (in percent)	...	...	...	26.5	26.9	26.5	16.8	26.9	26.6	27.5	23.6	**...**	**26.0**
Government revenues (excluding grants, in percent of GDP)	12.9	12.7	13.2	13.4	13.9	14.4	14.9	15.3	15.7	17.6	17.8	**12.5**	**15.7**
Aid flows (in Billion of US dollars) 5/	0.2	0.3	0.2	0.4	0.5	0.5	0.5	0.5	0.5	0.5	0.6		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	2.2	2.3	2.1	1.9	1.7	1.6	1.3	0.9	**...**	**1.7**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	36.8	41.7	42.4	28.6	45.5	45.9	49.9	54.1	**...**	**43.9**
Nominal GDP (Billion of US dollars)	14	16	18	18	20	22	24	26	28	42	87		
Nominal dollar GDP growth	0.9	8.9	12.9	4.4	8.9	9.0	9.1	8.9	8.5	8.0	7.5	**5.5**	**8.1**
Memorandum items:													
PV of external debt 7/	...	...	28.4	30.7	30.8	30.0	31.2	29.8	28.6	20.4	10.9		
In percent of exports	...	...	118.6	121.3	129.2	130.4	136.9	131.9	127.4	90.8	48.6		
Total external debt service-to-exports ratio	4.9	5.5	8.5	7.0	8.0	10.8	10.9	10.4	9.2	10.9	4.8		
PV of PPG external debt (in Billion of US dollars)			5.0	5.7	6.2	6.6	7.5	7.7	8.1	8.5	9.5		
(PVt-PVt-1)/GDPt-1 (in percent)				3.6	2.9	1.9	4.0	1.2	1.3	-0.1	0.2		
Non-interest current account deficit that stabilizes debt ratio	-2.3	-1.9	-4.4	4.4	4.7	4.6	3.1	5.3	4.7	5.5	4.6		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 2. Benin: Public Sector Debt Sustainability Framework, Baseline Scenario, 2019-2042

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
Public sector debt 1/	**41.2**	**46.1**	**49.9**	**51.6**	**52.0**	**50.7**	**49.2**	**47.9**	**46.9**	**43.6**	**44.1**	**34.5**	**47.4**
of which: external debt	**25.3**	**28.4**	**36.5**	**37.5**	**37.6**	**36.8**	**37.8**	**36.2**	**34.7**	**25.1**	**13.6**	**19.5**	**33.3**
Change in public sector debt	0.1	4.9	3.8	1.6	0.5	-1.4	-1.5	-1.2	-1.1	-0.4	0.3		
Identified debt-creating flows	**-1.4**	**1.4**	**3.9**	**0.9**	**0.3**	**-0.9**	**-0.8**	**-0.7**	**-0.6**	**-0.3**	**0.3**	**1.5**	**-0.4**
Primary deficit	**-1.1**	**2.7**	**3.5**	**3.7**	**2.8**	**1.3**	**1.3**	**1.3**	**1.4**	**1.3**	**1.3**	**2.0**	**1.7**
Revenue and grants	14.1	14.4	14.1	14.3	15.0	15.5	15.9	16.2	16.6	18.4	18.4	**13.4**	**16.6**
of which: grants	1.2	1.7	0.9	0.8	1.1	1.1	1.0	1.0	0.9	0.8	0.6		
Primary (noninterest) expenditure	13.0	17.1	17.6	17.9	17.8	16.8	17.2	17.5	18.0	19.7	19.7	**15.3**	**18.3**
Automatic debt dynamics	**-0.4**	**-3.3**	**0.5**	**-3.4**	**-2.5**	**-2.2**	**-2.2**	**-2.1**	**-2.0**	**-1.6**	**-0.9**	**-0.7**	**-2.1**
Contribution from interest rate/growth differential	-1.2	-0.8	-1.9	-3.4	-2.5	-2.2	-2.2	-2.1	-2.0	-1.6	-0.9	**-1.0**	**-2.1**
of which: contribution from average real interest rate	1.5	0.7	1.1	-0.7	0.5	0.7	0.7	0.7	0.7	0.9	1.3	**0.7**	**0.6**
of which: contribution from real GDP growth	-2.6	-1.5	-3.1	-2.7	-3.0	-2.9	-2.9	-2.8	-2.7	-2.4	-2.2	**-1.6**	**-2.7**
Contribution from real exchange rate depreciation	0.8	-2.4	2.4	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**1.9**	**-0.1**	**0.7**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.2**	**0.1**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	2.0	0.0	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**1.6**	**3.5**	**-0.1**	**0.7**	**0.2**	**-0.5**	**-0.7**	**-0.5**	**-0.4**	**-0.1**	**0.0**	**1.3**	**-0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**43.2**	**44.1**	**44.5**	**43.3**	**42.0**	**41.1**	**40.4**	**38.7**	**41.4**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**305.7**	**309.0**	**296.6**	**280.1**	**264.8**	**253.6**	**244.1**	**210.4**	**225.0**		
Debt service-to-revenue and grants ratio 3/	**45.8**	**42.6**	**76.7**	**32.5**	**35.8**	**33.0**	**37.9**	**29.5**	**35.3**	**36.0**	**45.5**		
Gross financing need 4/	5.4	10.8	14.2	9.0	8.2	6.4	7.4	6.1	7.2	7.9	9.6		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	6.9	3.8	7.2	5.7	6.2	6.0	6.0	6.0	6.0	5.9	5.4	**5.4**	**6.0**
Average nominal interest rate on external debt (in percent)	2.4	2.1	2.8	2.5	2.5	2.5	2.5	2.7	2.7	2.7	2.4	**1.9**	**2.6**
Average real interest rate on domestic debt (in percent)	6.5	3.5	7.2	-0.4	3.9	4.0	3.6	3.9	3.6	4.2	4.4	**3.5**	**3.6**
Real exchange rate depreciation (in percent, + indicates depreciation)	4.3	-9.9	9.1	...	...	...	...	...	...	...	...	**2.0**	**...**
Inflation rate (GDP deflator, in percent)	-0.3	2.9	1.6	4.8	1.0	1.4	2.0	2.0	2.0	2.0	2.0	**1.5**	**2.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	-6.6	36.7	10.3	7.6	5.5	-0.2	8.8	8.2	8.4	5.9	5.4	**8.5**	**7.1**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-1.2	-2.2	-0.3	2.0	2.3	2.7	2.8	2.6	2.4	1.7	1.0	**-1.2**	**2.2**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

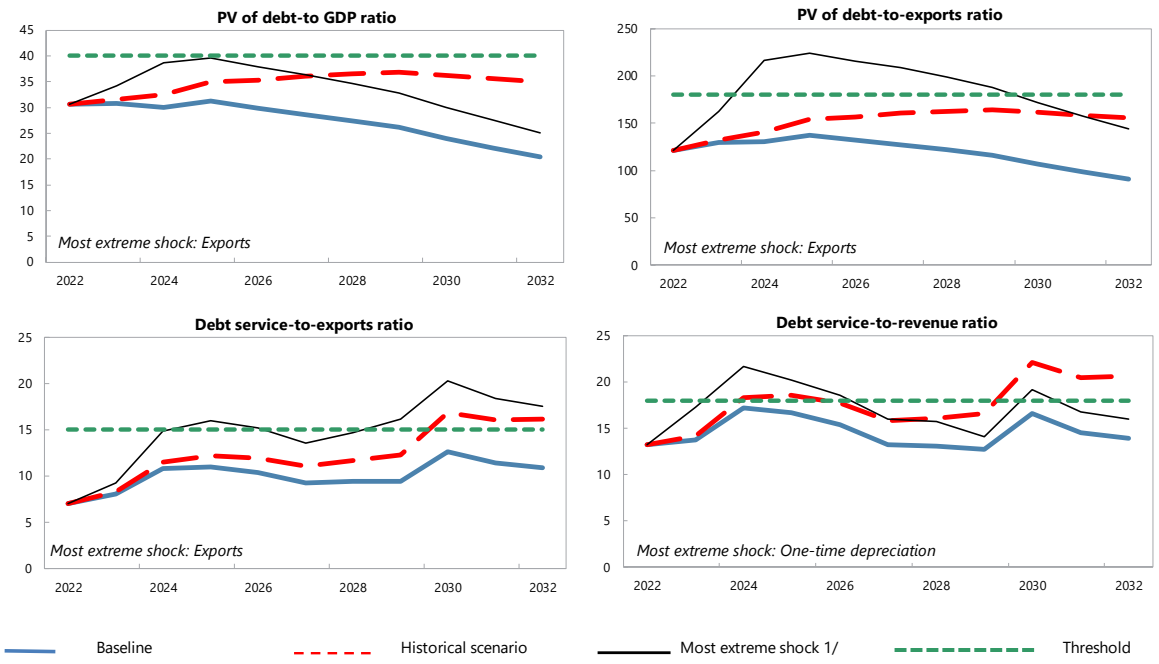
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Benin: Indicators of Public Guaranteed External Debt under Alternatives Scenarios. 2022-2032[2]



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	No	No
Commodity price	No	No
Market financing	Yes	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.5%	2.9%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	20	18
Avg. grace period	6	4

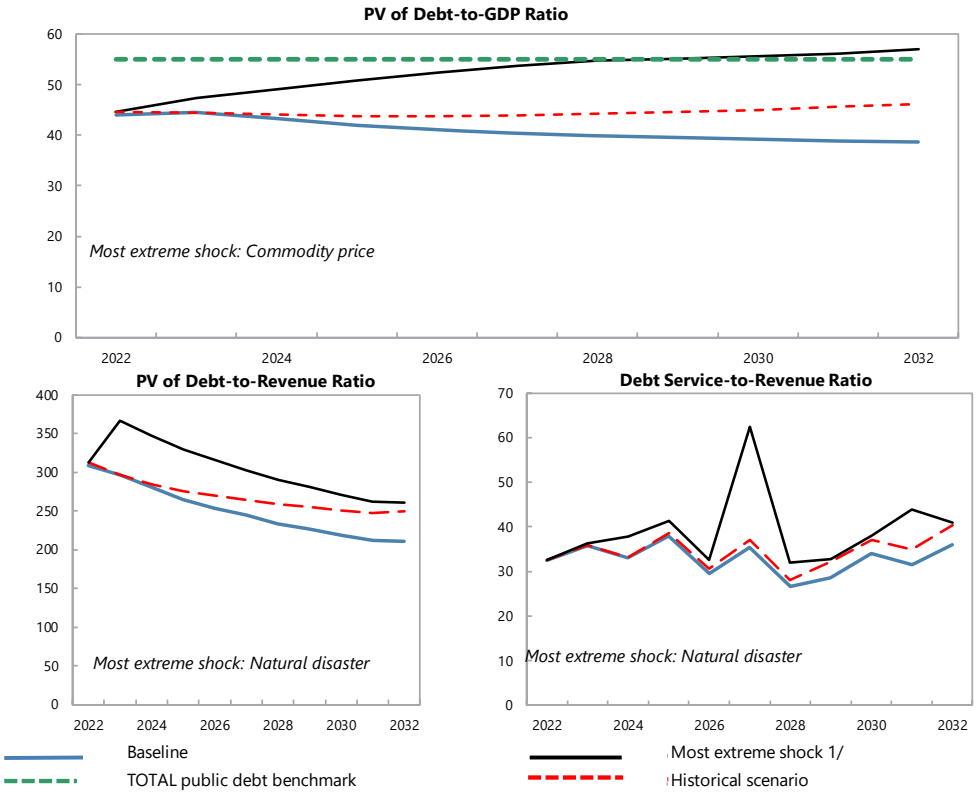
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most extreme shock even after disregarding the one-ff breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF Research Department.

Figure 2. Benin: Indicators of Public Debt Under Alternatives Scenarios, 2022-2032



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	44%	44%
Domestic medium and long-term	53%	53%
Domestic short-term	3%	3%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.5%	2.9%
Avg. maturity (incl. grace period)	20	18
Avg. grace period	6	4
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.2%	4.2%
Avg. maturity (incl. grace period)	4	4
Avg. grace period	3	3
Domestic short-term debt		
Avg. real interest rate	3.4%	3.4%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Benin: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2022-2032

(Percent of GDP)

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of debt-to GDP ratio											
Baseline	31	31	30	31	30	29	27	26	24	22	20
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	31	32	32	35	35	36	37	37	36	36	35
B. Bound Tests											
B1. Real GDP growth	31	32	33	34	32	31	30	28	26	24	22
B2. Primary balance	31	32	32	33	31	31	29	28	26	24	22
B3. Exports	31	34	39	40	38	36	35	33	30	27	25
B4. Other flows 3/	31	33	33	34	33	32	30	29	26	24	22
B5. Depreciation	31	39	34	36	34	33	31	30	28	26	24
B6. Combination of B1-B5	31	36	35	36	34	33	31	30	27	25	23
C. Tailored Tests											
C1. Combined contingent liabilities	31	34	33	34	33	33	32	30	28	27	25
C2. Natural disaster	31	35	34	35	34	34	33	32	30	28	27
C3. Commodity price	31	35	38	40	38	36	35	32	29	27	24
C4. Market Financing	31	34	33	35	33	32	31	29	27	25	23
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	121	129	130	137	132	127	122	116	107	99	91
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	121	133	141	154	156	161	163	164	161	159	155
B. Bound Tests											
B1. Real GDP growth	121	129	130	137	132	127	122	116	107	99	91
B2. Primary balance	121	132	139	145	139	136	131	125	115	107	99
B3. Exports	121	163	**217**	**224**	**216**	**209**	**199**	**188**	172	158	144
B4. Other flows 3/	121	136	145	151	146	141	134	128	117	108	99
B5. Depreciation	121	129	117	124	120	116	111	106	98	91	84
B6. Combination of B1-B5	121	149	136	164	158	152	145	138	127	117	108
C. Tailored Tests											
C1. Combined contingent liabilities	121	143	144	151	145	146	141	135	125	118	110
C2. Natural disaster	121	150	152	159	154	156	150	145	135	129	121
C3. Commodity price	121	**185**	**202**	**202**	**188**	175	160	150	136	123	112
C4. Market Financing	121	129	130	137	132	128	123	117	107	98	90
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	7	8	11	11	10	9	9	9	13	11	11
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	7	8	11	12	12	11	12	12	**17**	**16**	**16**
B. Bound Tests											
B1. Real GDP growth	7	8	11	11	10	9	9	9	13	11	11
B2. Primary balance	7	8	11	11	11	10	10	10	13	12	12
B3. Exports	7	9	15	**16**	**15**	14	15	**16**	**20**	**18**	**17**
B4. Other flows 3/	7	8	11	11	11	10	10	11	14	12	12
B5. Depreciation	7	8	11	11	10	9	9	8	12	10	10
B6. Combination of B1-B5	7	9	13	13	12	11	12	11	**15**	14	13
C. Tailored Tests											
C1. Combined contingent liabilities	7	8	11	11	11	10	10	10	13	12	11
C2. Natural disaster	7	8	12	12	11	10	10	10	14	13	12
C3. Commodity price	7	10	14	14	13	11	12	13	**16**	15	14
C4. Market Financing	7	8	11	11	11	10	10	13	**16**	11	11
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	13	14	17	17	15	13	13	13	17	15	14
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	13	14	**18**	**19**	18	16	16	17	**22**	**20**	**21**
B. Bound Tests											
B1. Real GDP growth	13	14	**19**	**18**	17	14	14	14	18	16	15
B2. Primary balance	13	14	18	17	16	14	14	14	18	15	15
B3. Exports	13	14	**18**	**19**	17	15	16	17	**21**	**18**	17
B4. Other flows 3/	13	14	18	17	16	14	14	14	**18**	16	15
B5. Depreciation	13	17	**22**	**20**	**19**	16	16	14	**19**	17	16
B6. Combination of B1-B5	13	15	**20**	**19**	17	15	16	15	**19**	17	16
C. Tailored Tests											
C1. Combined contingent liabilities	13	14	18	17	16	14	14	13	17	15	15
C2. Natural disaster	13	14	**18**	18	16	14	14	14	18	15	15
C3. Commodity price	13	16	**21**	**21**	**19**	16	17	17	**21**	**18**	17
C4. Market Financing	13	14	17	17	16	14	13	18	**21**	14	14
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

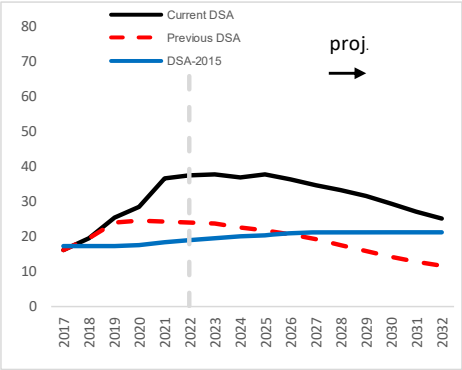
Table 4. Benin: Sensitivity Analysis for Key Indicators of Public Debt, 2022-2032

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of Debt-to-GDP Ratio											
Baseline	44	45	43	42	41	40	40	40	39	39	39
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	45	45	44	44	44	44	44	45	45	46	46
B. Bound Tests											
B1. Real GDP growth	45	48	49	49	49	49	50	50	51	52	53
B2. Primary balance	45	47	48	46	45	44	44	43	42	42	42
B3. Exports	44	47	51	49	48	47	46	45	44	43	43
B4. Other flows 3/	44	46	47	45	44	43	43	42	41	41	40
B5. Depreciation	45	52	49	46	44	42	40	38	37	35	34
B6. Combination of B1-B5	45	45	45	43	42	42	41	40	40	39	39
C. Tailored Tests											
C1. Combined contingent liabilities	45	53	52	50	49	48	47	46	46	45	45
C2. Natural disaster	45	**55**	54	52	51	50	50	49	49	48	48
C3. Commodity price	45	47	49	51	52	54	55	**55**	**56**	**56**	**57**
C4. Market Financing	44	45	43	42	41	41	40	40	39	39	39
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	309	297	280	265	254	244	234	226	219	211	210
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	313	297	285	275	269	264	258	255	251	247	250
B. Bound Tests											
B1. Real GDP growth	313	316	316	306	300	295	289	287	283	280	285
B2. Primary balance	313	311	308	291	278	267	255	246	237	229	227
B3. Exports	309	314	327	309	295	283	269	258	246	235	231
B4. Other flows 3/	309	308	302	285	273	262	250	241	231	222	220
B5. Depreciation	313	351	321	293	272	253	235	220	206	192	185
B6. Combination of B1-B5	313	298	291	274	262	251	239	230	221	213	211
C. Tailored Tests											
C1. Combined contingent liabilities	313	354	334	315	302	288	275	265	255	245	243
C2. Natural disaster	313	367	347	329	316	303	290	281	271	262	260
C3. Commodity price	313	352	352	355	348	339	326	315	310	305	309
C4. Market Financing	309	297	280	265	254	245	235	227	218	211	210
Debt Service-to-Revenue Ratio											
Baseline	32	36	33	38	29	35	27	29	34	32	36
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	32	36	33	39	31	37	28	32	37	35	40
B. Bound Tests											
B1. Real GDP growth	32	37	36	42	33	41	33	36	42	41	47
B2. Primary balance	32	36	34	40	31	41	33	31	36	35	40
B3. Exports	32	36	34	39	31	37	29	32	37	34	39
B4. Other flows 3/	32	36	33	39	30	36	28	30	36	33	37
B5. Depreciation	32	36	37	41	33	37	29	31	37	34	37
B6. Combination of B1-B5	32	35	33	39	30	36	29	29	34	31	37
C. Tailored Tests											
C1. Combined contingent liabilities	32	36	37	40	32	58	30	31	36	41	39
C2. Natural disaster	32	36	38	41	33	62	32	33	38	44	41
C3. Commodity price	32	40	38	45	35	42	33	42	47	44	49
C4. Market Financing	32	36	33	38	30	36	27	33	38	31	36

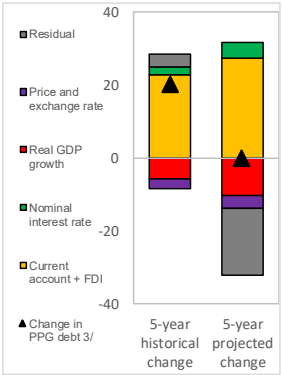
Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 3. Benin: Drivers of Debt Dynamics – Baseline Scenario

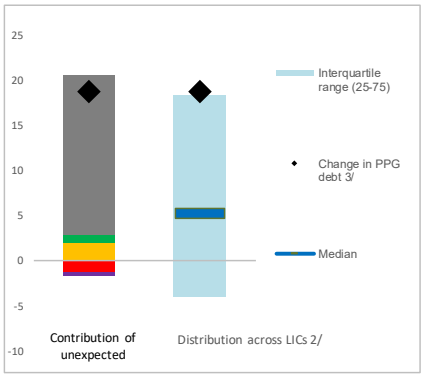
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

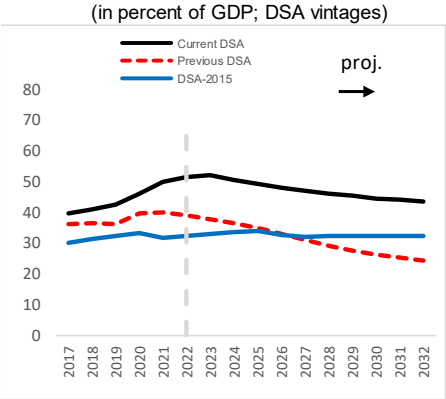


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

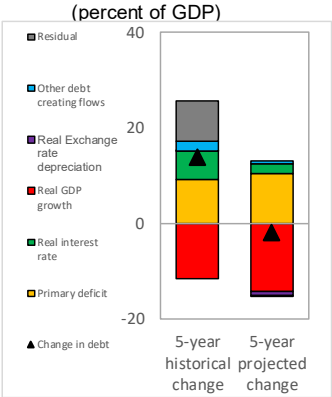


Public debt

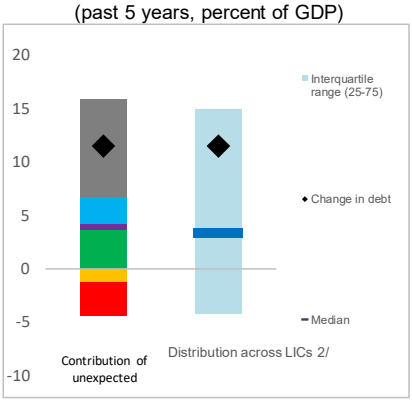
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



Sources: Country authorities; and staff estimates and projections.

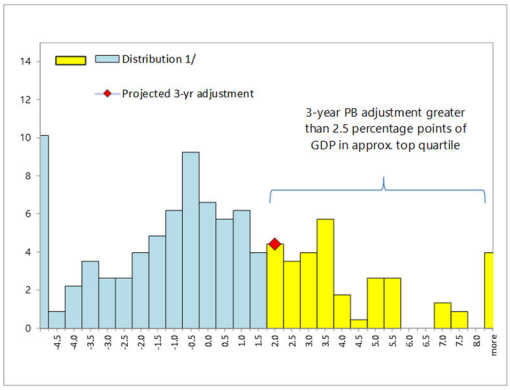
1/ Difference between anticipated and actual contributions on debt rations.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

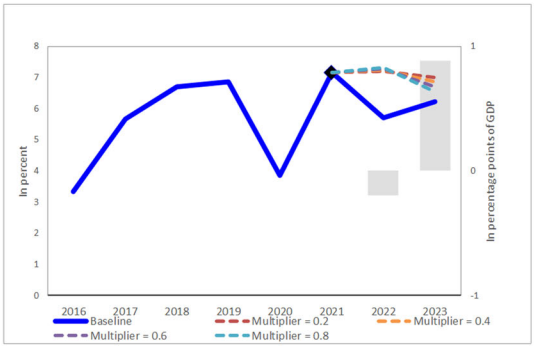
Figure 4. Benin: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



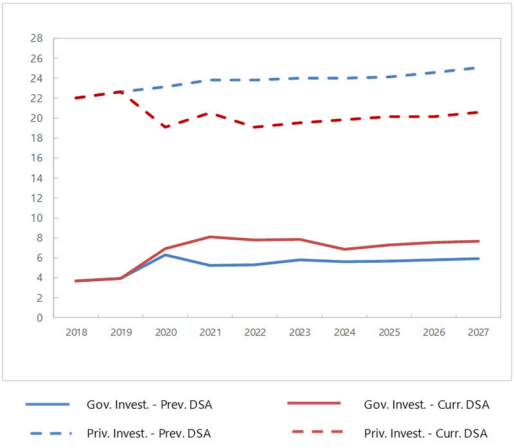
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

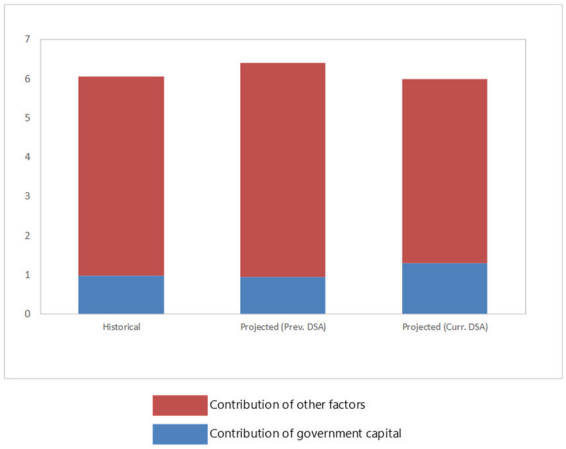


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

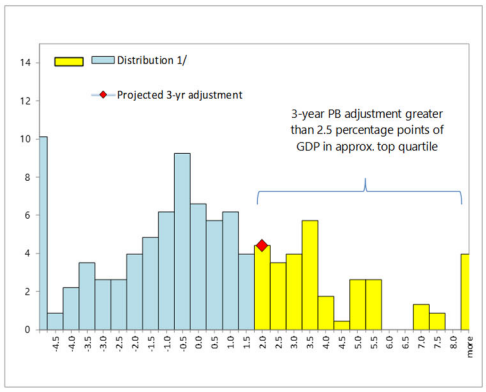


Sources: Country authorities; and staff estimates and projections

Figure 4. Benin: Realism Tools

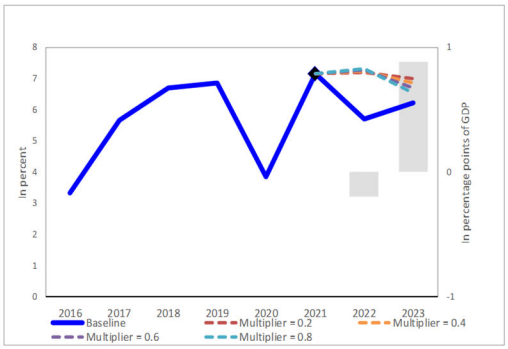
3-year Adjustment in Primary Balance

(Percentage Point of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

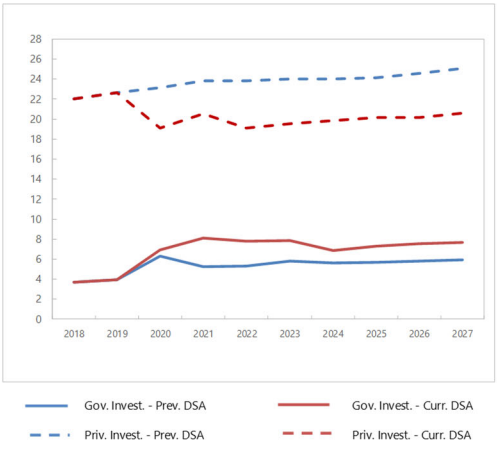
Fiscal Adjustment and Possible Growth Path[1/]



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

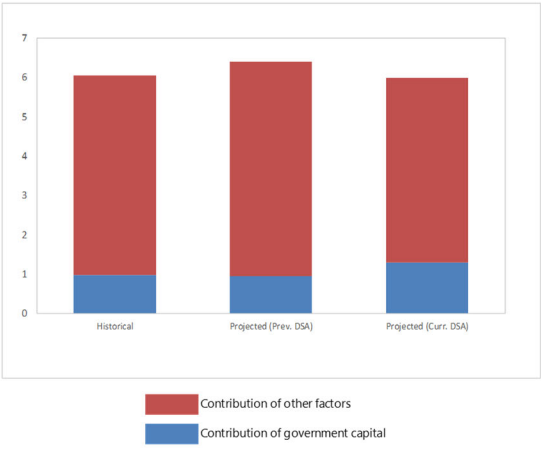
Public and Private Investment Rates

(Percent of GDP)



Contribution to Real GDP Growth

(Percent 5-year Average)



Sources: Country authorities; and staff estimates and projections

Figure 5. Benin: Qualification of the Moderate Category, 2022-2032 1/

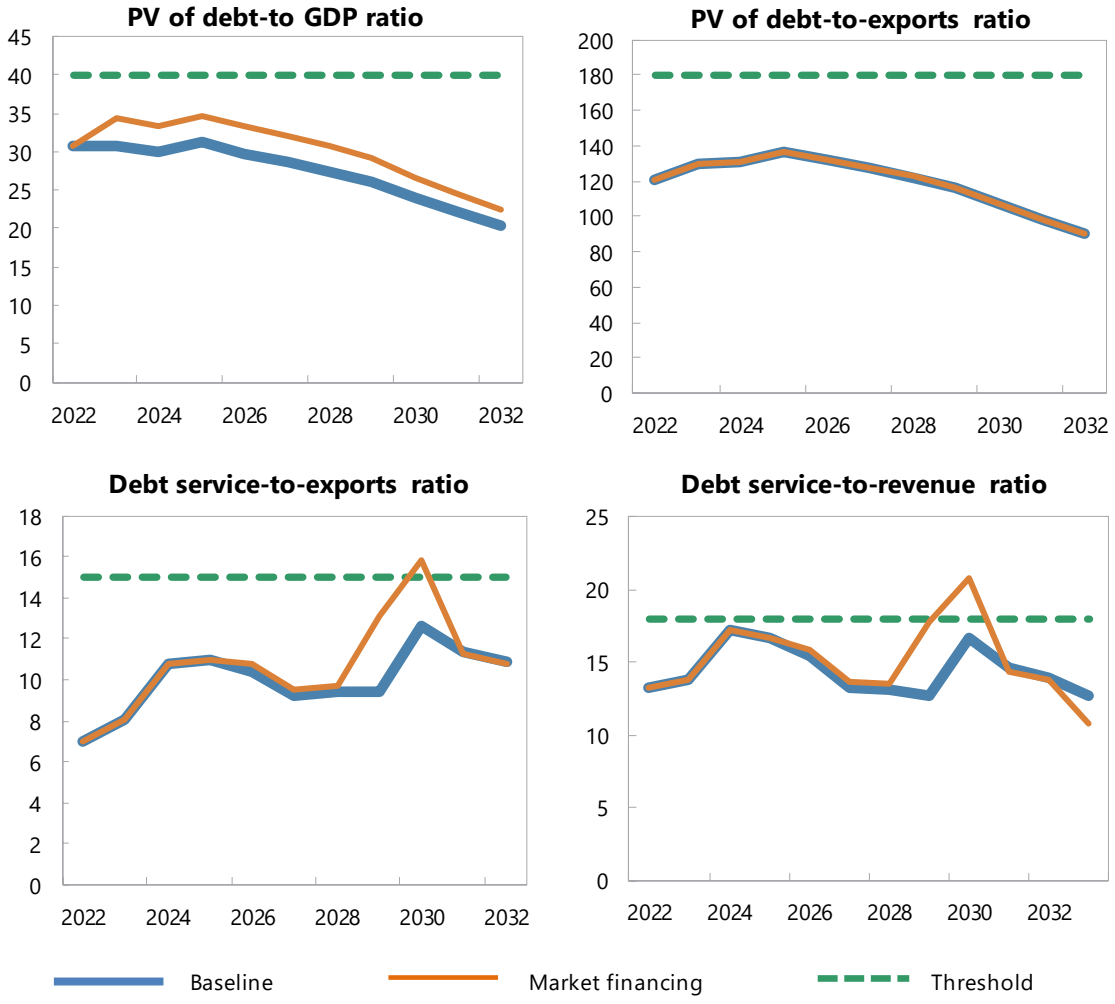
Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

Figure 6. Benin: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	9	677
Breach of benchmark	No	Yes
Potential heightened liquidity needs	Moderate	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spread as of May 3, 2022



Sources: Country authorities; and staff estimates and projections.